SOLUTIONS, SUSPENSIONS, COLLOIDS, EMULSIONS 5

Solutions, suspensions, colloids and emulsions are all mixtures; they often involve particles scattered through a liquid.

A **solution** has particles (solute) mixed evenly throughout the medium (solvent). Although the dissolved particles can colour the solution, the individual solute particles cannot be seen. When a solution is passed through filter paper, the solute particles are small enough to fit through the tiny pores.

1 Write the missing word: 'When you mix a solid with a liquid and the solid disappears, the solid is said to have _____.'

2 Most of the solutions used in the laboratory are *aqueous*. Explain what is meant by aqueous.

3 Solutions are often described as *homogeneous*. Explain what is meant by homogeneous.

4 A student added copper sulfate to water and stirred it until no solid pieces could be seen. He said, 'The solid can't have dissolved because the mixture is coloured.' Discuss his statement.

5 What is meant by a *saturated solution*?

When a liquid can mix with water to form a solution, it is said to be **miscible** with water. (Liquids are only said to be miscible with other liquids, not solids or gases.) Ethanol (C_2H_5OH) and water are miscible.

6 What do you think two liquids that do not mix properly to form a solution are called?

A **suspension** has particles suspended in a medium. These particles do not all separate to mix evenly throughout the medium; they stay in clumps large enough to be seen with the naked eye. Suspensions tend to be cloudy until they settle out to from a sediment at the bottom of the container. The particles in suspensions do not pass through filter paper.

7 Give an example of a suspension.

8 Explain what is meant by 'a suspension settles out'.

A **colloid** is a type of suspension, but the particles do not settle out. The particles cannot be seen with the naked eye but can be seen through a microscope. The particles cannot pass through filter paper. Smoke in air and natural rubber are colloids. Sols are types of colloids; they are made up of very fine particles spread evenly throughout a fluid. An aerosol is a colloid where the liquid or solid particles are spread through a gas. The word *aerosol* is often used to describe the containers (spray cans or vaporisers) that produce an aerosol. Gels are colloids where the liquid medium is viscous and forms a jelly-like substance.

9 Name three substances that are packaged in aerosol containers.

10 Fog is similar to smoke; outline the similarities and differences between these two substances.

11 Name a type of gel and outline its use.

12 The term *medium* is often used in reference to mixtures. What is a medium?

13 Give a new example of a colloid.

The Tyndall effect

John Tyndall (1820–1893) was a British scientist who studied the scattering of light by dust and other small particles. This scattering of light is now called the **Tyndall effect**. If a beam of light is shone through a clear fluid in which small particles are suspended, the longer wavelengths of visible light (e.g. blue) are scattered less effectively than the shorter wavelengths of light (e.g. red). True solutions do not scatter light.

14 Explain whether the Tyndall effect would be shown in each substance.

a fog

b sugar dissolved in water

Research

15 Glass is a *super-cooled solution*. Find out what this means.

16 Robert Brown was one of the first people to study colloids and the first to describe a phenomenon now called Brownian motion. Outline what Brownian motion involves.

ISBN 978 1 92561 066 6
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